De La Salle Holy Cross College

NOVEMBER EXAMINATIONS

2024

Grade 10

MATHEMATICS

Paper 2

Examiner:	J.P. Guilhermino	**Time:**	2 hours
Moderators:	G. De Freitas and A. Seewald	**Marks:**	110

Name:	Marking Guidelines (post marking)		
Teacher:	Mrs De Freitas	Mr Guilhermino	Mrs Seewald

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 20 pages, 10 questions, extra space and a formula sheet on page 2. Please check that your question paper is complete.
2. Answer the questions on the question paper and hand it in at the end of the examination. Remember to write your name in the space provided.
3. Non-programmable and non-graphical calculators may be used, unless otherwise indicated. **Ensure that it is set in DEGREE mode**.
4. All necessary calculations must be clearly shown; answers only do not guarantee full marks.
5. Diagrams and graphs are not ne → See annotations: Q2, Q4 and Q7
6. Round off final answers to **ONE** decimal place if necessary, unless otherwise specified.
7. It is in your own interest to write legibly and to present your work neatly.
8. One blank page (page 20) is included in the paper. If you run out of space for an answer, then use this page. Clearly indicate the question number of your answer should you use this extra space.

Geo Trig Midpt Trig△ Graph A-G Mean Box W Data Geo

	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Q 9	Q 10	Total	
Mark	△	○	○	○	○	△		PG				
Total	6	11	5	10	14	24	10	6	9	15	110	%

FORMULA SHEET

$$T_n = a + d(n-1)$$

$$A = P(1+in)$$

$$A = P(1+i)^n$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$\overline{x} = \frac{\Sigma f \cdot x}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

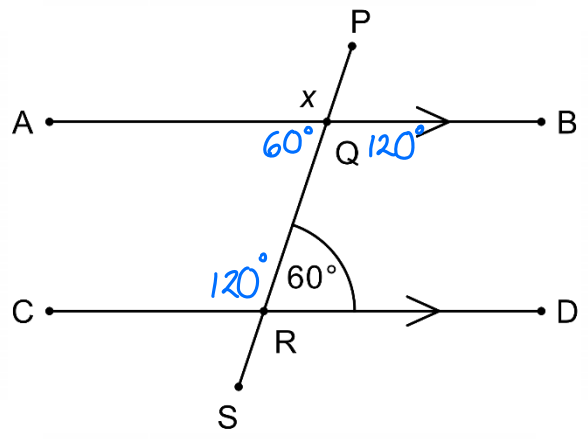
$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

QUESTION 1

Based on the given information for each scenario, write down the value of x in the table. No reasons or working out are necessary.

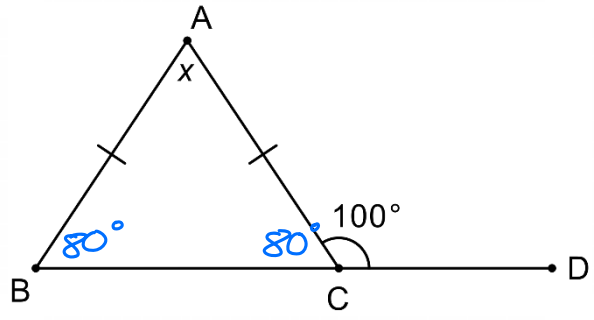
QUESTION	(a)	(b)	(c)
ANSWER	$x =$ 120° a ✓✓ (2)	$x =$ 20° a ✓✓ (2)	$x =$ 25° a ✓✓ (2)

(a)



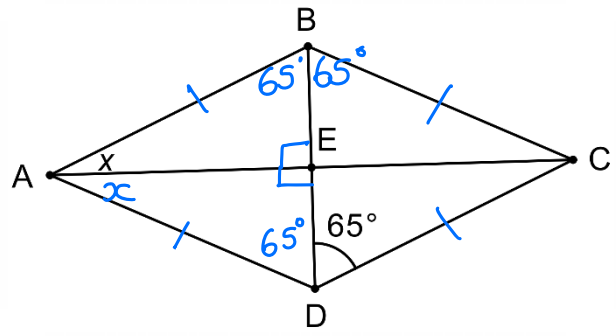
AB || CD, $P\hat{R}D = 60°$ and $A\hat{Q}P = x$.
PS is a straight line.

(b)



AB = AC; $A\hat{C}D = 100°$ and $\hat{A} = x$.
BD is a straight line.

(c)



ABCD is a rhombus and its diagonals intersect at E.
$E\hat{D}C = 65°$ and $B\hat{A}E = x$.

[6]

QUESTION 2

(a) Use a calculator to write down the following values given that $\alpha = 67{,}5°$ and $\beta = 100°$. Round off your answers to 3 decimal digits.

(1) $\tan^2(\alpha + \beta) =$ $(\tan 167{,}5°)^2 = 0{,}049148\ldots$ (2)

$(-0{,}2216\ldots)^2 \approx 0{,}049$ ✓

□ outside or $-0{,}22\ldots$

only ≈ final (if shown working) * P ≈ 3 dp

(full marks ans only) (½ if no working and wrong ans.)

(2) $3.\sin\alpha + \sin 2\beta =$ (2)

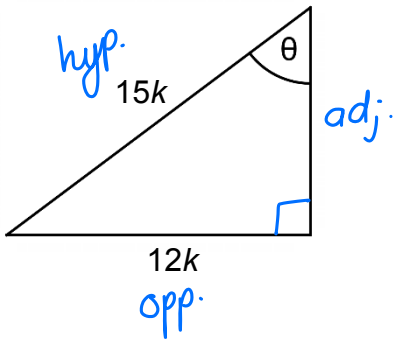
$= 3\sin 67{,}5° + \sin 200°$

$= 2{,}77\ldots + (-0{,}34\ldots)$ → (½ for intermediate values)

$= 2{,}42961\ldots \approx 2{,}43$ ✓✓

(b) Determine the size of θ in the right-angled triangle below. (2)

1 dp (pen. ≈ here)



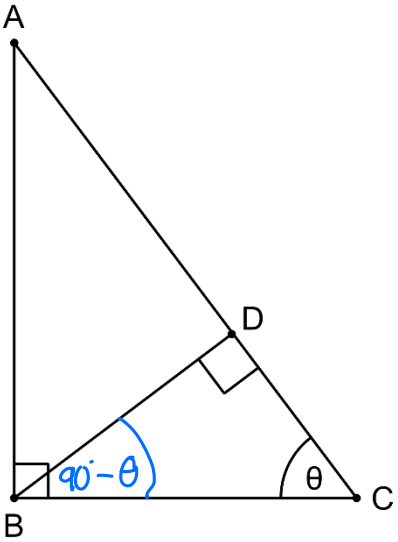
$\sin\theta = \frac{12k}{15k}$ ✓

$\theta = \sin^{-1}\left(\frac{4}{5}\right)$

$= 53{,}130\ldots°$

$\approx 53{,}1°$ ✓ ca ratio

(c) Use the diagram to fill in the missing side or trig. ratio. (2)



$\tan\theta = \frac{AB}{BC}$ ✓a

$\cos(90° - \theta) = \frac{BD}{BC}$ ✓a

(d) If $\tan A = 1{,}05$, then calculate the value of $2\sin(2A - 40°)$ correct to 3 decimal places. Show necessary working. (3)

(p ≈ 3dp)

$\hat{A} = \tan^{-1}(1{,}05)$

$= 46{,}397\ldots°$ ✓

$2 \times \sin(2 \times 46{,}397\ldots° - 40°)$ ✓

$= 2\sin(52{,}79436\ldots°)$

$= 1{,}59294\ldots$

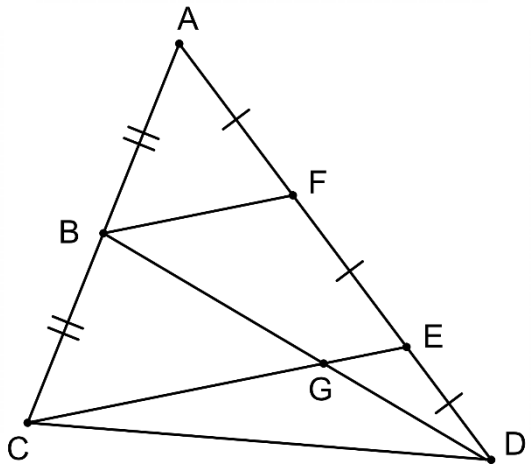
$\approx 1{,}593$ ✓ (only ≈ at end)

[11]

QUESTION 3

AB = BC, GE = 15 units and AF = FE = ED.

The proof below will calculate CG. Complete the proof by filling in the missing statements and reasons.

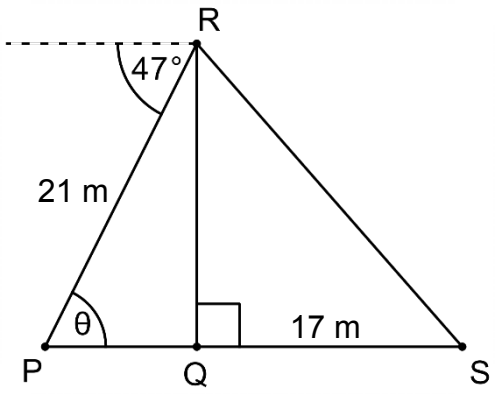


Statement	**Reason**
In △ACE:	
AB = BC and AF = FE	Given
∴ BF // CE ✓a and CE = 2.BF	mid-pt thrm. ✓a
In △FBD:	
FE = ED and BF || GE	
∴ BF = 2·GE ✓a (= 30 u) or (CE = 2·BF = 60 u)	mid-pt. thrm converse ✓a
so CG = 45 ✓ca units	

[5]

QUESTION 4 (P. ≈ 1dp)

RQ is a vertical pole. The foot of the pole, Q, is on the same horizontal plane as P and S. The pole is anchored with cables RS and RP. The angle of depression from the top of the pole to P is 47°. PR is 21 m and QS is 17 m. $\hat{RPQ} = \theta$.



(a) Write down the size of θ. Give a reason. (1)

$\theta = 47°$ ✓a ; alt. ∠s ✓R (// lines) not pen.

(b) Calculate RQ. (3)

$\sin 47° = \frac{RQ}{21}$ ✓

$\Rightarrow RQ = 21 \sin 47° = 15,358\ldots \approx 15,4\text{ m}$ ✓

✓

(c) Hence, calculate the size of $\hat{S}$. (2)

$\tan \hat{S} = \frac{15,4}{17}$ ✓

$\Rightarrow \hat{S} = \tan^{-1}\left(\frac{15,4}{17}\right) = 42,1728\ldots° \approx 42,2°$ ✓ ca; ratio

(d) If P, Q and S lie in a straight line, then calculate how far apart the base of the cables are. (4)

$\cos 47° = \frac{PQ}{21}$ ✓* $\Rightarrow PQ = 21 \cos 47° = 14,3219\ldots\text{ m}$ ✓* ✓*

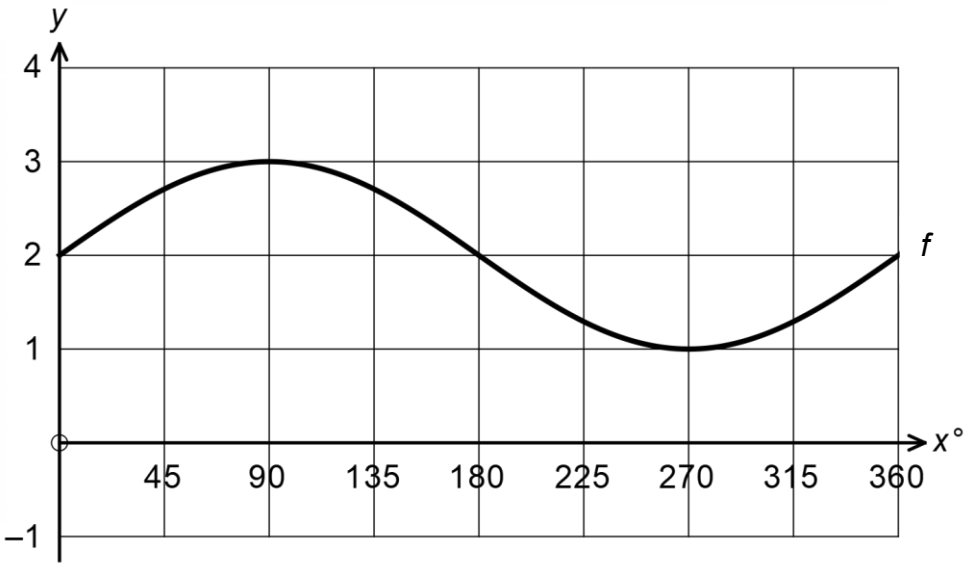
$PS = 17 + PQ \approx 31,3\text{ m}$ ✓

or Pythag. in △RPQ
✓S
✓R (with reason and
✓+✓ ca Q(b) → RQ)

[10]

QUESTION 5

The graph of $f(x)$ is drawn for $x \in [0° ; 360°]$.



(a) Write down the period of f. (1)

360° ✓a

(b) Write down the amplitude of f. (1)

1 unit ✓a

(c) Determine the equation of f. (3)

$f(x) = 1 \cdot \sin x + 2$

✓"a" ✓ratio ✓"q"

(d) For which values of x is f decreasing? (2)

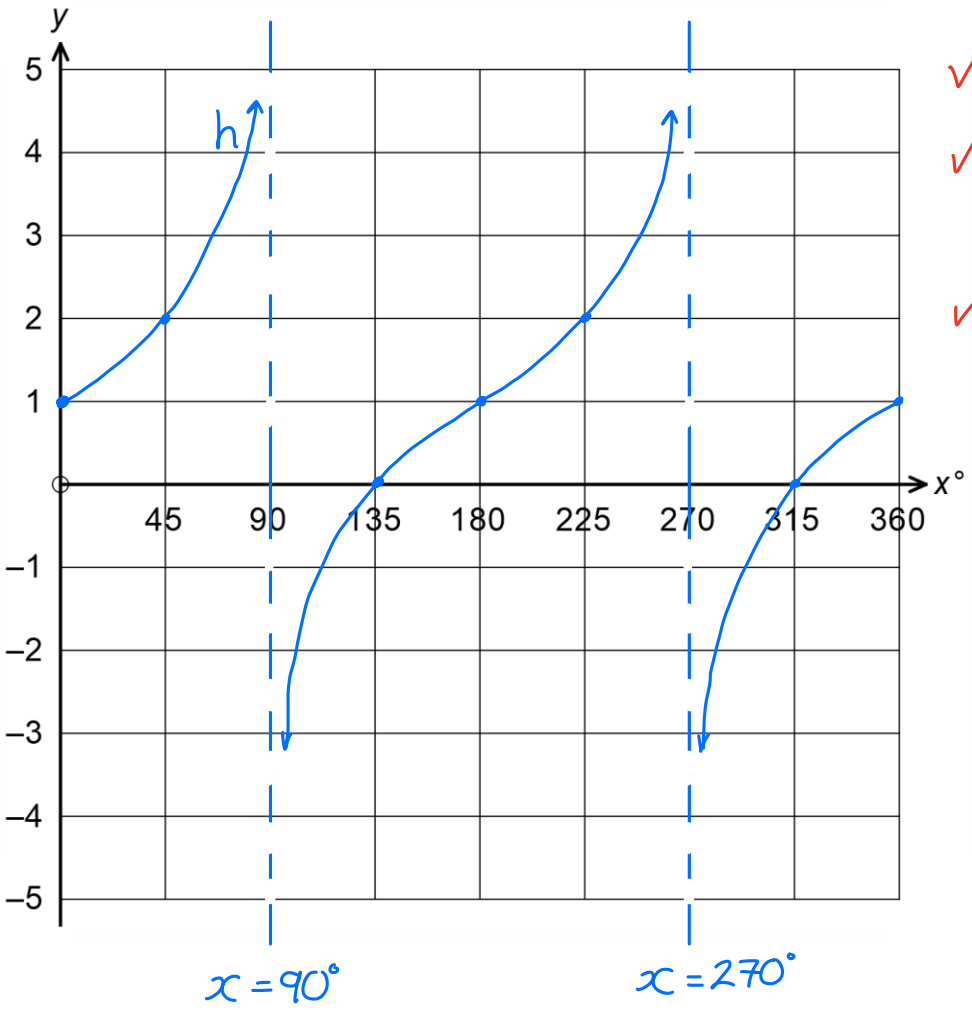
$x \in (90°; 270°)$ or $90° < x < 270°$

✓ ✓

(e) Use the graph to determine the value(s) of x when $f(x) = 2$. (2)

$x = \{0°; 180°; 360°\}$ ✓✓a -1: per error (omission/incorrect)

(f) On the axes below, draw sketch graph of $h(x) = \tan x + 1$ for $x \in [0° ; 360°]$. Clearly label any asymptotes and/or turning points as well as intercepts with the axes. (3)



✓asymp's with equns
✓shp (tan x up l4)
(-1 for wrong y-int)
✓x-ints

(g) Determine the equation of $m(x)$ if m is created by reflecting $h(x)$ over the x-axis and then shifting it 1 unit downwards. (2)

$m(x) = -h(x) - 1$

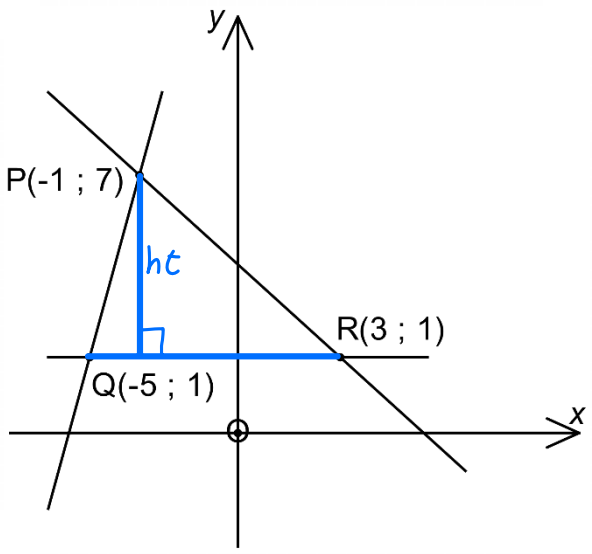
$= -\tan x - 2$

✓"a" ✓"q"

[14]

QUESTION 6

P(-1 ; 7), Q(-5 ; 1) and R(3 ; 1) are the vertices of a triangle in the Cartesian plane.



(a) Write down the value of QR. (1)

$QR = 8$ units ✓a

(b) Calculate PQ and leave your answer in surd form. (2)

$PQ = \sqrt{(7-1)^2+(-1+5)^2}$ ✓m

$= \sqrt{36+16}$

$= \sqrt{52}$ or $2\sqrt{13}$ u ✓a

ca

(c) Show that △PQR is isosceles. (2)

$PR = \sqrt{(7-1)^2+(-1-3)^2} = \sqrt{36+16} = \sqrt{52}$ or $2\sqrt{13}$ u ✓m ✓a

$\therefore PR = PQ \neq QR$ ✓

$(\therefore \triangle PQR$ is isosceles$)$

(d) Determine whether △PQR has a right angle at P. Give an analytical reason. (4)

- $m_{PQ} = \frac{7-1}{-1+5} = \frac{6}{4}$ ✓ma and $m_{PR} = \frac{7-1}{-1-3} = -\frac{6}{4}$ ✓ma
- $\therefore m_{PQ} \times m_{PR} \neq -1$ ✓ca (Reason)

$\therefore$ PQ $\not\perp$ PR ✓ca

or Accept: Pythag. converse (LHS ... RHS) with prev.-calc'd lengths

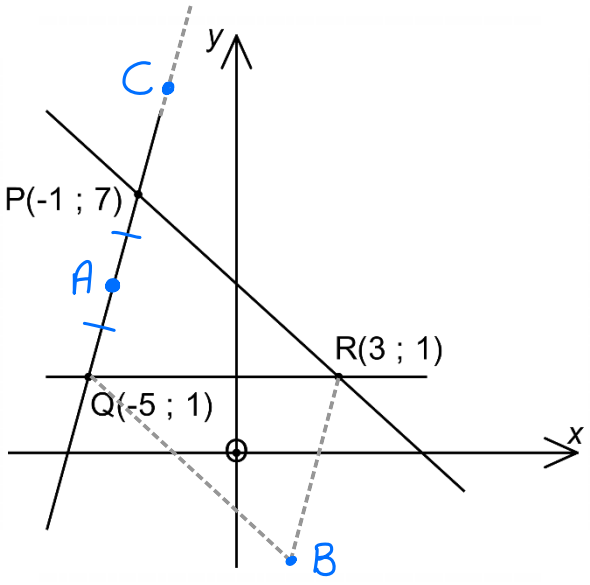
Not accept: trig. ratios (90° △ not given)

(e) Calculate the area of △PQR. (3)

$A_{\triangle PQR} = \frac{1}{2} \cdot 8 \cdot 6 = 24\ u^2$ ✓base ✓ht ✓ma

ca : Qa

* If 90° found at P, then ca PQ and PR as b/h dim's.



(f) Determine the coordinates of A if A is a point on PQ such that PA = AQ. (2)

$A\left(\frac{-1-5}{2}\ ;\ \frac{7+1}{2}\right) = A(-3\ ;\ 4)$ ✓a ✓a

$\frac{0}{2}$ if no brackets

(g) B is a point such that PQBR is a parallelogram.

(1) Write down the coordinates B. (2)

$P \to R : (x\ ;\ y) \to (x+4\ ;\ y-6)$ so $Q \to B : B(-1\ ;\ -5)$ ✓a ✓a

or $P \to Q : (x\ ;\ y) \to (x-4\ ;\ y-6)$ so $R \to B$

(2) What type of parallelogram will PQBR be? Give a geometric reason. (2)

Rhombus, parm. with adj. sides = ✓a ✓R

(ca prev. pg : isos △ and 90°)

(h) P, Q and C(a ; 9) are collinear points. Calculate a. (4)

- $m_{PQ} = \frac{3}{2}$ (from QD) ✓

- $\frac{3}{2} = \frac{2}{a+1} \Rightarrow a+1 = \frac{4}{3} \Rightarrow a = \frac{1}{3}$ ✓ ca (solve)

✓ equn* ✓ m_{PS} or* $\frac{2}{a+1} = \frac{8}{a+5} \Rightarrow (a+5) = 4(a+1)$

or $m_{QS} = \frac{8}{a+5}$ $\Rightarrow 1 = 3a$

$\Rightarrow a = \frac{1}{3}$

(i) Write down the equation of the line that passes through the midpoints of PQ and PR. No reason is required. (2)

$y = 4$ (Mid-pt Thrm.)

✓ horz. ✓ val. → ca ht. in QE

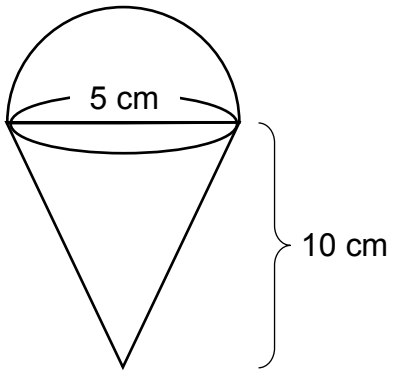
[24]

QUESTION 7

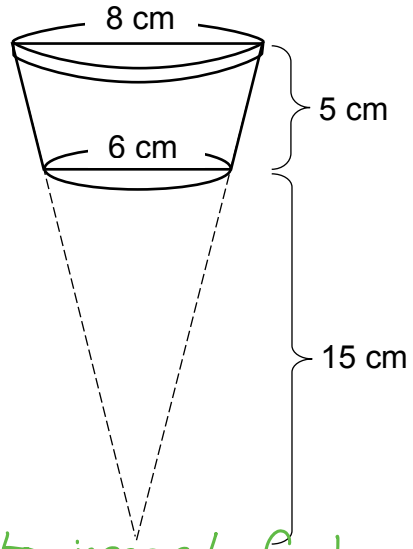
You may find the following useful when answering this question.	
Volume of a sphere = $\frac{4}{3}\pi r^3$	Surface area of a sphere = $4\pi r^2$
Volume of a cone = $\frac{1}{3}\pi r^2 h$	Surface area of a cone = $\pi \cdot r^2 + \pi \cdot r \cdot$ slant height

Delicious Dairy offers two options for their ice-creams, shown below.
Option 1 is a cone filled with ice-cream and topped with a scoop in the shape of a hemisphere.
Option 2 is a cup in the shape of a *truncated* cone. The top diameter is 8 cm and the bottom diameter is 6 cm. A *truncated* cone is a cone that has had its point cut off parallel to its base.

Option 1

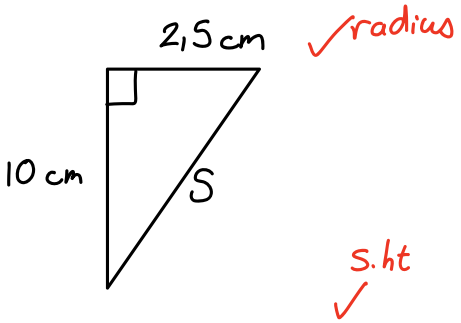


Option 2



↗ P ≈ 1 dp or intermediate ≈ leading to incorrect final ans.

(a) Option 1 comes in a cardboard holder that neatly fits around the cone. Calculate the surface area of the cardboard (ignore flaps). (3)



✓radius

✓s.ht

$S = \frac{5\sqrt{17}}{2}$ cm

$= 10,3077\ldots$ cm

(Pythag.)

↳ Do not pen. reason

$SA = \pi(2,5)\left(\frac{5\sqrt{17}}{2}\right)$

$= 80,9569\ldots$ cm²

≈ 81 cm² ✓ma (adj fmla)

Not p. ≈

(b) Calculate which option holds the greater volume of ice-cream. Show all working out. (7)

#1: $V_{tot} = V_{hemisphere} + V_{cone}$

$= \frac{1}{2} \cdot \frac{4}{3}\pi(2,5)^3 + \frac{1}{3}\pi(2,5)^2(10)$ ✓ma ✓ma ✓ma

$\frac{1}{2}\left(\frac{125}{6}\pi\right) = 32,724\ldots$ $\quad \frac{125}{6}\pi = 65,449\ldots$

$= 31\frac{1}{4}\pi$ or $\frac{125}{4}\pi$ cm^3

$= 98,174\ldots\ cm^3$

#2: $V_{tot} = V_{big\ cone} - V_{small\ cone}$

$= \frac{1}{3}\pi(4)^2(20) - \frac{\pi}{3}(3)^2(15)$ ✓ma ✓ma ✓ma

$106\frac{2}{3}\pi$ $\quad \frac{320}{3}\pi$ $\quad 335,103\ldots$ $\quad 45\pi = 141,37\ldots$

$= 61\frac{2}{3}\pi$ or $\frac{185}{3}\pi$ cm^3

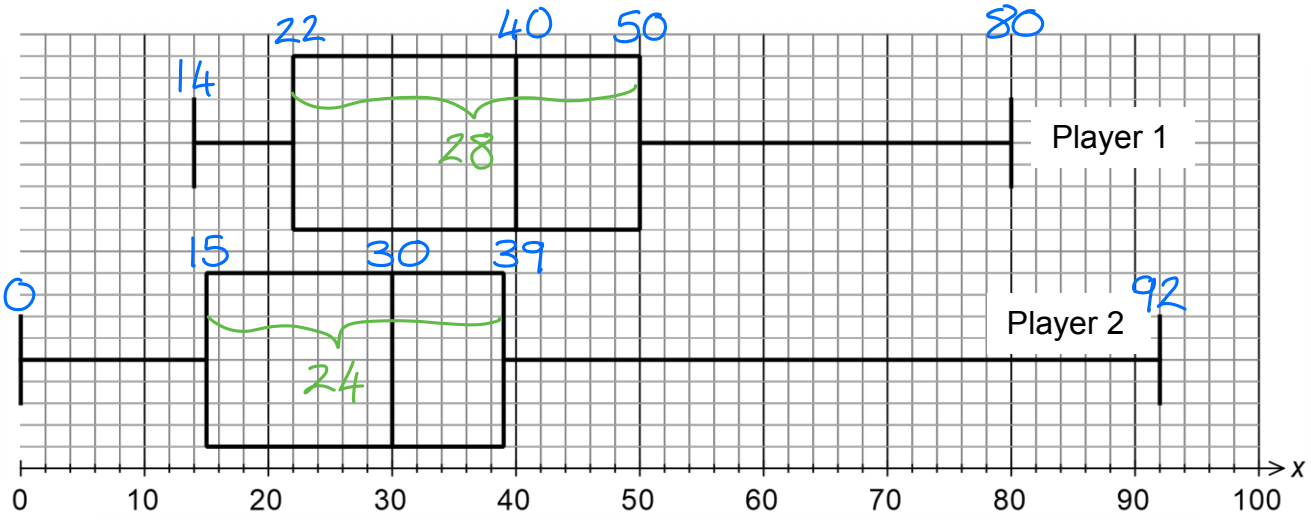
[10]

$= 193,731\ldots\ cm^3$

∴ Option 2 holds more ice-cream. ✓ca

QUESTION 8

Two Cricketers are being considered for selection based on their runs during the past season.
A higher number of runs is better than a low number of runs.



(a) Determine the inter-quartile range for player 1. (2)

$50 - 22 = 28$

✓values ✓ma

(b) Player 2 claims that they make more than x runs 75% of the time.
What is the value of x? (1)

$x = 15$ ✓a

(c) Assuming both players played a similar number of matches this season, which player would you select for the team?
Refer to <u>two</u> statistical concepts to support your choice. (3)

Player 1 – lower range or more consistant
or less spread-out

✓a: Player 1

✓R ✓R

- Q_2 (Player 1) > Q_3 (Player 2)
- higher min.
- almost 100% of Player 1 vals > 75% P2's vals
- Q_2 (P1) > Q_2 (P2)
- P1 skewed right (so expect lower values) **[6]**

QUESTION 9

For a project on climate change a pupil measured the temperature every day at the same time. The data is summarised below.

Temperature (°C)	Frequency
$20 \leq T < 24$	2
$24 \leq T < 28$	4
$28 \leq T < 32$	8
$32 \leq T < 36$	7
$36 \leq T < 40$	6
$40 \leq T < 44$	3

(a) How many days did the pupil collect data for? (1)

$n = 30$ days ✓a

(b) Write down the modal class for the data. (1)

$28 \leq T < 32$ ✓a

(c) In which class interval does the lower quartile (Q_1) lie? (1)

Pos $Q_1 = \frac{1}{4}(31) = 7,75^{th}$ → between 7th & 8th

Q_1 is in $28 \leq T < 32$ ✓a (ca: if show calc. for position)

(d) The estimated mean for the data was calculated as $\frac{98}{3}$. Unfortunately the pupil discovered that all the data in the last class interval (shaded) was recorded incorrectly. The pupil then moved all of the data from the last class interval to the same correct class interval. The correct estimated mean for the data is $\frac{466}{15}$.

Determine which class interval the data was moved to. (6)

See pg. 20 (more likely sol.)

854

- Sum of all temp's (with wrong data) $= \frac{98}{3} \times 30 = 980$
- Sum of all correct data $= \frac{466}{15} \times 30 = 932$

✓ma

- Wrong data $= 3 \times 42^{\circ}C = 126$ * ✓mid pts

✓Σ of all others

- $932 - (980 - 126) = 78$

✓Σ of data pts in correct class *

- $78 \div 3$ data pts $= 26^{\circ}C$ ✓m

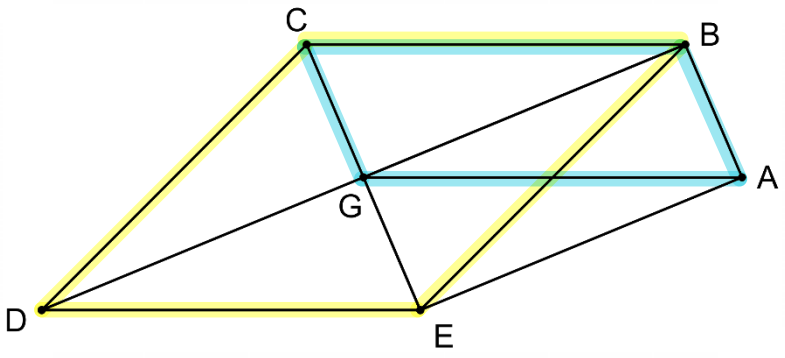
∴ Moved to $24 \leq T < 28$ ✓ca **[9]**

QUESTION 10 a a a ✓✓✓

(a) Write down three ways to prove that a quadrilateral is a parallelogram. (3)

- ① Both pairs opp. sides ∥.
  ② Both pairs opp. sides =.
- ③ Both pairs opp. ∠s =.
  ④ One pair opp. sides = and ∥.
- ⑤ Diag's. bisect each other.

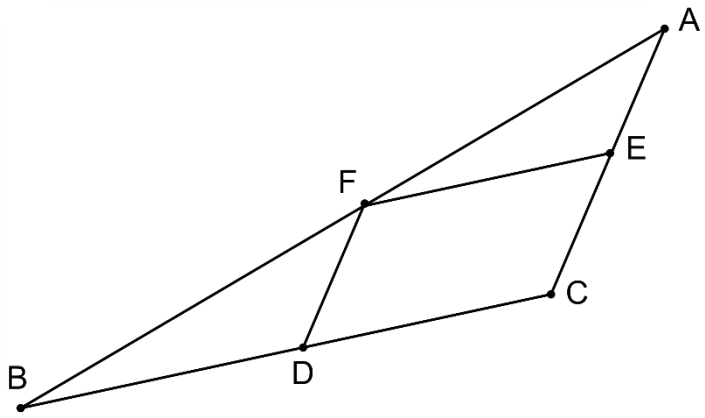
(b) BCDE and ABCG are parallelograms.
Prove that ABGE is a parallelogram. Give reasons.



Statement	Reason	(5)
$BA = CG$	, opp. sides of parm. ABCG	✓S+R
$CG = GE$	, diag's. of parm. BCDE	✓S+R
$\therefore BA = GE$		✓∴
$BA \parallel CGE$	, opp. sides of parm. ABCG	✓S+R
$\therefore$ ABGE is a parm., quad. with 1 pair opp. sides = and ∥.		✓R

(c) FDCE is a parallelogram. CE is produced to A such that CE = EA and CD is produced to B such that CD = BD.

Prove △BDF ≡ △FEA. Give reasons.



Statement		**Reason**	(7)
$AE = EC$	✓S+R	, given	
$FD = EC$	✓S+R	, opp. sides of parm.	
$\therefore AE = FD$	✓∴		
$BD = DC$	✓	, given	or, similarly $BD = FE$
$FE = DC$		, opp. sides of parm.	
$\therefore BD = FE$			

Opt. ①: $FE \parallel BC$, opp. sides of parm
and $AE = EC$
$\therefore AF = FB$, mid-pt thm. converse ✓S ✓R
$\therefore \triangle BDF \equiv \triangle FEA$, SSS ✓∴

Opt. ②: $FE \parallel BC$ and $FD \parallel EC$, opp. sides of parm.
✓ $\hat{C} = A\hat{E}F$, corresp. ∠s $FE \parallel BC$
$\hat{C} = F\hat{D}B$, corresp. ∠s $FD \parallel EC$
✓ $\therefore A\hat{E}F = F\hat{D}B$
✓∴ $\therefore \triangle BDF \equiv \triangle FEA$, SAS

[15]

GRAND TOTAL: 110

ADDITIONAL SPACE (ALL QUESTIONS)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED.

Q9.d

Σ of all other correct data ✓ 854 ✓ mid-pts

$$\frac{466}{15} = \frac{2(22) + 4(26) + 8(30) + 7(34) + 6(38) + 3x}{30}$$

✓ x 30

$$\Rightarrow 932 = 3x + 854$$

$$\Rightarrow 3x = 78 \quad \checkmark \Sigma \text{ of new values}$$

$$\Rightarrow x = 26 \quad \checkmark \div 3$$

$$\therefore \text{Moved to } 24 \leq T < 28 \quad \checkmark \text{ca}$$